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Efficiency = work performed / heat absorbed

$$\eta = \frac{|q_h| - |q_c|}{|q_h|}$$

q_h - heat absorbed from the hot reservoir

q_c - heat transferred to cold reservoir

$$\eta = 1 - \frac{|q_c|}{|q_h|} \quad ---(7.27)$$

For a reversible cyclic process

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings} = 0$$

$$\Delta S_{system} = - \Delta S_{surroundings}$$

$$\frac{q_h}{T_h} = \frac{-q_c}{T_c}$$

$$\frac{-T_c}{T_h} = \frac{q_c}{q_h}$$

$$\frac{T_c}{T_h} = \frac{|q_c|}{|q_h|} ----7.28$$

Sustituting 7.28 in 7.27

$$\Rightarrow \eta = 1 - \frac{T_c}{T_h} ------7.29$$

$T_h >> T_c$

Hence, $\eta < 1$

efficiency in percentage can be expressed as

$$\text{Efficiency in percentage} = \left[1 - \frac{T_c}{T_h}\right] x\,100$$

Clausius statement:

It is impossible to transfer heat from a cold reservoir to a hot reservoir without doing some work.

Problem: 7.10

If an automobile engine burns petrol at a temperature of 816° C and if the surrounding temperature is 21° C, calculate its maximum possible efficiency.

Solution:

$$\% \text{ Efficiency} = \left[\frac{T_h - T_c}{T_h}\right] \times 100$$

Here

T_h = 816+273= 1089 K;

T_c= 21+273= 294K

$$\% \text{ Efficiency} = \left(\frac{1089 - 294}{1089}\right) \times 100$$

% Efficiency=73%

Evaluate Yourself - 5

An engine operating between 127^0 C and 47^0 C takes some specified amount of heat from a high temperature reservoir. Assuming that there are no frictional losses, calculate the percentage efficiency of an engine.

Unit of entropy:

The entropy (S) is equal to heat energy exchanged (q) divided by the temperature (T) at which the exchange takes place. Therefore, The SI unit of entropy is JK^{-1}.

Spontaneity and Randomness

Careful examination shows that in each of the processes viz., melting of ice and evaporation of water, there is an increase in randomness or disorder of the system. The water molecules in ice